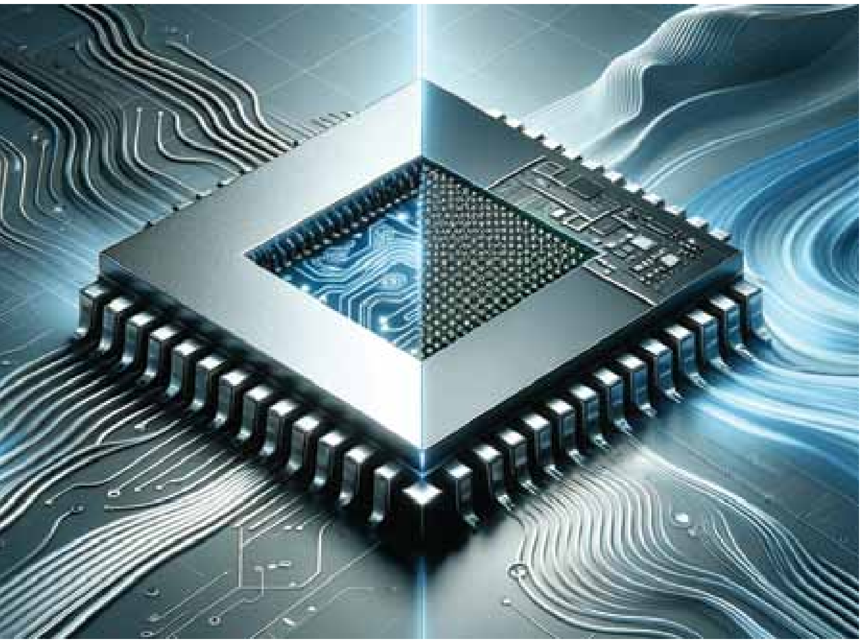
RESEARCH
& INNOVATION UPDATES

Analogue chips to offer digital precision with analogue efficiency and speed



Memristor system-on-chip to power up the analogue world with digital precision (Courtesy: Tetramem Inc.)

Researchers at USC Viterbi School of Engineering have advanced analogue computing, potentially revolutionising data processing in autonomous vehicles and AI. Their paper in Science introduces a design for analogue chips combining digital precision with analogue efficiency and speed, focusing on memristors for data storage and processing. Led by Prof. J. Joshua Yang, the team modified memristors for unparalleled precision in analogue computing. This breakthrough extends to other memory technologies like magnetic memories and phase-change memories. Yang's lab developed new circuit architecture and algorithms for precise, rapid programming of analogue devices, enhancing neural network training efficiency and speed. This innovation opens new avenues beyond AI and ML, such as scientific computing for weather forecasting, marking a significant step in merging digital and analogue systems for a more efficient computing future.

New tech turns cars into mobile health diagnostics centres for drivers

Inspired by Star Trek's USS Enterprise medical bay, researchers at the University of Waterloo have developed a radar technology for vehicle health monitoring, turning cars into mobile health centres. This technology aims to collect health data during commutes, eliminating the need for wearables. The compact radar, smaller than a USB thumb drive, is integrated into the vehicle's interior, detecting body vibrations, and processing them with AI to create a medical profile. Drivers receive health reports on their cell phones after their journey. The system can detect subtle movements like breathing and heartbeats, and it has been validated for respiratory and heart conditions. The team plans to enhance the technology to monitor all occupants' health, run diagnostics, and facilitate emergency communication.

Radar sensor held up next to 10-cent coin to give size comparison (Credit: University of Waterloo)